A CLOSED IMMERSION IS OF FINITE PRESENTATION EXACTLY WHEN ITS SHEAF OF IDEALS IS OF FINITE TYPE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Statement span 4206-4216 and proof span 4218-4227 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The statement carries an author slogan ('Closed immersions of finite presentation correspond to quasi-coherent sheaves of ideals of finite type.'); the project's own shared preamble defines the slogan environment as a comment environment, so it is not typeset, and it is preserved verbatim, as part of the counted statement span.

Editorial note. The proof's closing phrase 'Hence we win.' is the author's and is preserved.

Editorial note. 1 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- On an affine open of the target a closed immersion is given by a quotient of the coordinate ring and its sheaf of ideals is the tilde of the corresponding ideal, cited from the schemes chapter. Used at proof lines 4218-4222. Reduces the statement to the affine case.
- A quasi-coherent module on an affine scheme is of finite type exactly when the corresponding module is finite, cited from the properties chapter. Cited at proof lines 4222-4224. Identifies finite typeness of the ideal sheaf with finiteness of the ideal in the coordinate ring.
- A quotient of a ring by an ideal is of finite presentation if and only if the ideal is a finite module. Used at proof lines 4225-4226. Matches the finite presentation condition on the closed immersion with the finiteness of the ideal.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01PB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2224–2224. Same-chapter reference used in the retained author text at source line(s) 703, 754, 4224, 5201, 5286, 6703, 10562, 11247. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *A closed immersion $i : Z \to X$ is of finite presentation if and only if the associated quasi-coherent sheaf of ideals $\mathcal{I} = \mathrm{Ker}(\mathcal{O}_X \to i_*\mathcal{O}_Z)$ is of finite type (as an $\mathcal{O}_X$-module).*

Proof. On any affine open $\mathrm{Spec}(R) \subset X$ we have $i^{-1}(\mathrm{Spec}(R)) = \mathrm{Spec}(R/I)$ and $\mathcal{I} = \widetilde{I}$. Moreover, $\mathcal{I}$ is of finite type if and only if I is a finite R-module for every such affine open (see Properties, Lemma 01PB). And R/I is of finite presentation over R if and only if I is a finite R-module. Hence we win. □